

AI 534 HW 3

1.

- 1.1. RSMLE can be defined as: a measure of the average difference between values predicted by a model and the actual observed values
- 1.2. RMSLE is similar to RMSE, but it looks at the errors in the logarithms of the predicted and actual values. This means it cares more about percentage differences than raw differences, so big numbers don't dominate the error as much.
- 1.3. This competition uses RMSLE instead because house prices can vary a lot. So it is used since it looks at errors in terms of percentages instead of actual dollar amounts. This way, getting a cheap house wrong by a bit counts about the same as getting an expensive house wrong proportionally. This makes it so that big houses don't dominate the score and smaller ones have a chance.
- 1.4. This means that on average the model predicted prices that are about 11% off from the actual prices.
- 1.5. My RMSLE error is .40613 and puts me at rank 5352.
- 1.6. We work in the log space because house prices can be all over the place, from small homes to really expensive ones. Using the log helps the model focus on percent mistakes instead of huge absolute differences, and it also makes extreme values less likely to throw things off.

2.

- 2.1. I get 7227 features
- 2.2.
 - 2.2.1. MSSubClass: 15
 - 2.2.2. MSZoning: 5
 - 2.2.3. LotFrontage: 108
 - 2.2.4. LotArea: 989
 - 2.2.5. Street: 2
 - 2.2.6. Alley: 3
 - 2.2.7. LotShape: 4
 - 2.2.8. LandContour: 4
 - 2.2.9. Utilities: 2
 - 2.2.10. LotConfig: 5
 - 2.2.11. LandSlope: 3
 - 2.2.12. Neighborhood: 25
 - 2.2.13. Condition1: 9
 - 2.2.14. Condition2: 8
 - 2.2.15. BldgType: 5
 - 2.2.16. HouseStyle: 8
 - 2.2.17. OverallQual: 10
 - 2.2.18. OverallCond: 9

2.2.19. YearBuilt: 110
2.2.20. YearRemodAdd: 61
2.2.21. RoofStyle: 6
2.2.22. RoofMatl: 8
2.2.23. Exterior1st: 15
2.2.24. Exterior2nd: 16
2.2.25. MasVnrType: 5
2.2.26. MasVnrArea: 305
2.2.27. ExterQual: 4
2.2.28. ExterCond: 5
2.2.29. Foundation: 6
2.2.30. BsmtQual: 5
2.2.31. BsmtCond: 5
2.2.32. BsmtExposure: 5
2.2.33. BsmtFinType1: 7
2.2.34. BsmtFinSF1: 601
2.2.35. BsmtFinType2: 7
2.2.36. BsmtFinSF2: 131
2.2.37. BsmtUnfSF: 730
2.2.38. TotalBsmtSF: 686
2.2.39. Heating: 6
2.2.40. HeatingQC: 4
2.2.41. CentralAir: 2
2.2.42. Electrical: 6
2.2.43. 1stFlrSF: 721
2.2.44. 2ndFlrSF: 390
2.2.45. LowQualFinSF: 21
2.2.46. GrLivArea: 810
2.2.47. BsmtFullBath: 4
2.2.48. BsmtHalfBath: 3
2.2.49. FullBath: 4
2.2.50. HalfBath: 3
2.2.51. BedroomAbvGr: 8
2.2.52. KitchenAbvGr: 4
2.2.53. KitchenQual: 4
2.2.54. TotRmsAbvGrd: 12
2.2.55. Functional: 7
2.2.56. Fireplaces: 4
2.2.57. FireplaceQu: 6
2.2.58. GarageType: 7
2.2.59. GarageYrBlt: 97
2.2.60. GarageFinish: 4
2.2.61. GarageCars: 5
2.2.62. GarageArea: 422

- 2.2.63. GarageQual: 6
- 2.2.64. GarageCond: 6
- 2.2.65. PavedDrive: 3
- 2.2.66. WoodDeckSF: 253
- 2.2.67. OpenPorchSF: 193
- 2.2.68. EnclosedPorch: 116
- 2.2.69. 3SsnPorch: 17
- 2.2.70. ScreenPorch: 72
- 2.2.71. PoolArea: 8
- 2.2.72. PoolQC: 4
- 2.2.73. Fence: 5
- 2.2.74. MiscFeature: 5
- 2.2.75. MiscVal: 21
- 2.2.76. MoSold: 12
- 2.2.77. YrSold: 5
- 2.2.78. SaleType: 9
- 2.2.79. SaleCondition: 6
- 2.3. My RMSLE on dev was .147

2.4.

Top 10 positive features:		
	feature	coef
5901	FullBath_3	35247.821051
1162	Neighborhood_StoneBr	32618.530855
1196	OverallQual_10	32397.599886
1155	Neighborhood_NoRidge	32256.344361
1398	RoofMatl_WdShngl	30932.458053
1204	OverallQual_9	29163.861723
462	LotArea_13891	22468.755471
6061	GarageCars_3	21541.919738
5921	TotRmsAbvGrd_10	21188.750215
6430	GarageArea_813	19328.798936

Top 10 negative features:		
	feature	coef
1959	BsmtFinSF1_2260	-15978.858085
664	LotArea_40094	-15978.858085
5791	GrLivArea_4676	-15978.858085
1710	MasVnrArea_762.0	-15978.858085
1198	OverallQual_3	-16119.367038
1199	OverallQual_4	-16215.610663
5927	TotRmsAbvGrd_4	-17116.026475
6059	GarageCars_1	-17947.447303
5899	FullBath_1	-18353.757753
15	MSZoning_C (all)	-19871.467086

Yes, I would say that this list of the top 10 positive and negative features do make sense to me. The top positive list is that having more full bathrooms, having a bigger garage, and a bigger lot area increase the value of the house. The top negative features show that a smaller lot area, less rooms, or garage storage also contribute to a less expensive house.

2.5.

No, you don't need to explicitly add a bias dimension like I needed to do in HW2. My feature weight is 253,065.52. This does make sense to me since this seems like a reasonable price for a house that doesn't have a lot going for it.

2.6

The meaning behind it is when all the features are set to 0. This is basically a reference house that is barebones with no added features that would add value to the house. It's the starting point of the house before the bells and whistles are added.

2.7

My score on kaggle was 0.21815. On my computer I got a value of .1707. I was ranked 4831 on this attempt.

3

3.1

Binarizing all features, even numeric ones, can make the data really sparse. This can make it ignore the actual size or order of numbers, and make it harder for the model to learn how things like lot size really affect house prices.

3.2

For mixed features like LotFrontage and GarageYrBlt, I'm keeping them as numeric and filling any missing values with the median using SimpleImputer. This lets the model still use the actual numbers while handling missing data without skewing the overall predictions.

3.3

- a. I get a total of 302 features.
- b. 0.1479 is my RMSLE

c.

Top 10 positive features:		
	feature	coef
79	Neighborhood_NoRidge	48515.009686
86	Neighborhood_StoneBr	44793.339511
132	RoofMatl_WdShngl	36814.423792
80	Neighborhood_NridgHt	29638.514985
273	PoolQC_Ex	23468.697975
155	Exterior2nd_ImStucc	20099.917040
70	Neighborhood_Crawfor	19930.392977
230	KitchenQual_Ex	19165.112674
183	BsmtQual_Ex	18021.390053
136	Exterior1st_BrkFace	17186.381072

Top 10 negative features:		
	feature	coef
50	LandContour_Bnk	-14373.441164
87	Neighborhood_Timber	-14812.205286
36	MSZoning_C (all)	-15995.971397
48	LotShape_IR3	-16216.806631
64	Neighborhood_Blmngtn	-16234.854639
71	Neighborhood_Edwards	-20801.555971
75	Neighborhood_Mitchel	-24460.366269
275	PoolQC_Gd	-26191.710265
102	Condition2_PosN	-28956.091009
125	RoofMatl_ClyTile	-34032.874011

Yes, these are different positive and negative features than I originally got.

- d. My kaggle score is .45763, and is still 0.1479 on dev.

4.

4.1 When I tried to use various different alpha values, my RMSLE got worse. It went from .1479 to .1614 with a $\alpha = .01$, then to .1641 at a $\alpha = 1$, then to .1611 at a $\alpha = 100$, then to .1614 at a $\alpha = 1000$.

4.2 This approach seemed to massively improve the model. I got a RMSLE value of .1393

4.3 Non-linear features let a linear model use relationships that aren't just straight lines. It allows it to pick up on interactions between variables or curves in the data. It's similar to XOR in perception in the way that without those extra features, the model won't really be very good. But if it does have them, it can solve more complex patterns.

4.4 I chose to combine the polynomialFeature and add alpha values to the pipeline. This approach gave me the best RMSLE value on my dev.

4.5 My best RMSLE on kaggle was .14244 and 0.1372 on my dev. This best submission is **test5.py**

2939

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0.14244

9

14s



Your Best Entry!

Your most recent submission scored 0.14244, which is an improvement over your previous score of 0.16260. Great job!

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5.

5.1 This project took me about 30 hours. I felt that this one took some more time to get into and understand. I felt like I needed to learn a lot to do it.

5.2 Id say this was on the more moderate to difficult scale for me

5.3 I worked on this alone

5.4 I feel at about 70% - 75% understanding. I still have something that I want to learn more about to feel like I have an understanding of the topic.

5.5 This was a neat assignment! I enjoyed working through it even though I found it more challenging then the past 2.